

# Science

Nature Notebook  
Nature Walks & Scouting  
Geology Lessons  
Geology Labs





## About the Course

This course includes the following topic(s): Geology Lessons, Geology Labs, Nature Notebook: Grades 9-12, Nature Walks & Scouting: Grades 9-12

**About Nature Notebook: Grades 9-12**

Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

**About Nature Walks & Scouting: Grades 9-12**

Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

**About Geology Lessons**

An elective in Earth Science, Geology is a wonderful course for Alveary high school students who are interested in the history, theory, and methods of the field.

**About Geology Labs**

Labs/fieldwork are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.



## Placement & Combining Tips

**Nature Notebook: Grades 9-12**

Learners may be combined and follow their own interests.

**Nature Walks & Scouting: Grades 9-12**

Learners may follow their own interests or follow the plan of their local scouting troop or natural history club.

**Geology Lessons**

Due to the complexity of certain topics and the maturity required, this course is recommended for Grades 11-12.



## Scheduling

| GRADE | SCHEDULE INFO.                                              | BOOKS                                                                                                                                                                                                                    |
|-------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9-12  | Nature Notebook: Grades 9-12<br>1+ time/week 20 min+        |                                                                                                                                                                                                                          |
| 9-12  | Nature Walks & Scouting: Grades 9-12<br>1 time/week 30 min+ |                                                                                                                                                                                                                          |
| 11-12 | Geology Lessons<br>5 times/week 45 min                      | 101 American Geo-Sites You've Gotta See<br>Annals of the Former World<br>Roadside Geology/Geology Underfoot Series<br>The Field Guide to Geology<br>The Map That Changed the World<br>The Story of the Earth in 25 Rocks |

| GRADE | SCHEDULE INFO.                     | BOOKS |
|-------|------------------------------------|-------|
| 11-12 | Geology Labs<br>1 time/week 60 min |       |

#### Sample Weekly View

| Day 1                                                      | Day 2           | Day 3           | Day 4                                              | Day 5                           |
|------------------------------------------------------------|-----------------|-----------------|----------------------------------------------------|---------------------------------|
| <b>Science: Geology</b>                                    |                 |                 |                                                    |                                 |
| Geology Lessons<br>Nature Walks &<br>Scouting: Grades 9-12 | Geology Lessons | Geology Lessons | Geology Lessons<br>Nature Notebook:<br>Grades 9-12 | Geology Lessons<br>Geology Labs |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

### Geology Lessons

- ☐ Carve out time to continue or establish the regular habit of spending time in nature, including the use of a nature journal, as appropriate.
- ☐ Obtain materials from the supply lists.
- ☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.
- ☐ Bookmark or print Quick Links, as needed.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Science: Geology

#### Geology Lessons



101 American Geo-Sites You've Gotta See



Annals of the Former World



Roadside Geology/Geology Underfoot Series



The Field Guide to Geology



The Map That Changed the World



## The Story of the Earth in 25 Rocks



### Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



### Quick Links

Science: Geology

#### Geology Lessons

- ∞ [Nature Journal Connection](#)
- ∞ [Alveary Bookshelf](#)
- ∞ [Roadside Geology](#)
- ∞ [State by State Rockhound](#)
- ∞ [How to Find Rocks](#)

Click THIS text  
or scan the QR  
code for links.



# Science: Geology

## How To Approach



### Introduce

- If starting a new book or a new topic in the book, then look at the title or a picture and take a moment to consider previous ideas and experiences.
- If continuing a previous reading, recap what was read previously. Often, the title of the book's section can help to draw out the main idea.



### Read

- Read or do, as instructed in the lessons, making note of or flagging unfamiliar terms, interesting ideas, important dates, inspiring quotes, etc.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.
- If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab/field work further supports major concepts from the text.



### Narrate

- Process the ideas of the lesson by retelling, defining a concept, explaining the links in a chain of thought, etc. Do this orally or silently to yourself.
- Use words, pictures, outlines, etc.
- If a particular idea cannot be narrated, then read or examine the text again.



### Discuss

- Consider with the teacher any thoughts, confusion, or concerns about the passage.
- If understanding is still uncertain, try rereading the passage or do some personal research on the topic.



## Connect

- Follow any extra links, examine any sidebars in the text, or pursue additional reading, depending on student interest.
-

# Science: Geology

## How To Approach



### Introduce

- Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.
- Relevant concept(s) are introduced in the text.
- Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.



### Lab Procedure

- Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.
- Record all data and observations in the lab notebook.



### Analysis & Conclusions

- After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.
- If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.



## Term 1

### WEEK 1 ☐ 45m Geology Lessons - Lesson 1

*How Geology Works*

☐ Materials: The Field Guide to Geology

#### → NOTE

If you have a rock and/or fossil kit, you may find it useful to keep it nearby throughout this course. As different materials are mentioned in your various books, you can reference these materials in your kit(s).

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 1 ☐ 45m Geology Lessons - Lesson 2

*Practical Interpretation*

☐ Materials: Roadside Geology & Notebook

#### → READ, NARRATE, & DISCUSS

Divide up your state or regional geology book so that you finish it this term. (If you are not able to finish it in one term, that is okay. You might choose to continue it next term or to leave off.) As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.

### WEEK 1 ☐ 45m Geology Lessons - Lesson 3

*Identification*

#### → INTRO

This is your day each week to prepare for your field studies using a mixture of resources provided for help, time for planning, and to spend time with the rocks you have found.

#### → RESEARCH

Start at Gator Girl Rocks to begin your journey. The How to Find Rocks website is helpful, as well. You will want to read about the laws that pertain to rockhounding and visit the links under your state. Read about any local resources, and begin making a plan. You might want to visit local nature centers and museums, and ask about local geology guides or guided tours. You might want to track down any local clubs or groups to join. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

∞ Link: Gator Girl Rocks

∞ Link: How to Find Rocks

∞ Link: Rock Clubs by State





## Term 1

### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 1 45m Geology Lessons - Lesson 4

#### *Geology Literature*

 Materials: Annals of the Former World

### → INTRO

People say this book is the "Literature of Geology". See what you think by the end of the term. McPhee travels across the country with various geologists, writing about the geological wonders along the way. He is writing for people who are not geologists, but that doesn't mean that the text is easy. Take in what you can and enjoy the views out the window.

This is a big book. Read Book 1 (Basin and Range), and then decide if you want to keep reading in order, read about places near you, or read about ones that you have never visited. Go at your own pace. If you read and narrate around 20 pages per week, you will get a good way through the book by the end of Term 2. Then you can decide if you want to finish as a free read or not.

### → READ, NARRATE, & DISCUSS

Be sure to look at the map on p.2-3 and read the Narrative Table of Contents before plunging into Chapter 1. Refer to the maps and charts in the text and at the end of the book while you are reading. You may want to look up unfamiliar terms as you read and make a collection of geological terms. Come up with an interesting way to narrate, perhaps an illustrated map or travel journal?

~ 20 pages

### WEEK 1 45m Geology Lessons - Lesson 5

#### *Flex Day/Current Events*

 Materials: Selected geology literature, lab/field supplies, or internet access

### → INTRO

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (linked in the lesson and in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

∞ Article Link: Geology Current Events



## Term 1

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| <p><b>WEEK 1</b> ☐ <b>60m Geology Labs - Lesson 1</b></p> <p>→ <b>OBSERVE</b></p> <p>Begin observations in the field: use your field notebook, collect specimens when allowed, and take pictures where helpful. Begin building a rock, mineral, and/or fossil collection.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                   |
| <p><b>WEEK 2</b> ☐ <b>45m Geology Lessons - Lesson 6</b></p> <p><i>How Geology Works</i></p> <div data-bbox="289 573 989 621" style="border: 1px solid #ccc; padding: 2px;"> <p>☐ Materials: The Field Guide to Geology</p> </div> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b></p> <p>Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.</p>                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>• <b>PLAN WEEKLY</b></p> <p>☐ nature walk - record observations</p> <p>☐ science free read</p> |
| <p><b>WEEK 2</b> ☐ <b>45m Geology Lessons - Lesson 7</b></p> <p><i>Practical Interpretation</i></p> <div data-bbox="289 867 989 915" style="border: 1px solid #ccc; padding: 2px;"> <p>☐ Materials: Roadside Geology &amp; Notebook</p> </div> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b></p> <p>Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.</p>                                                                                                                                                                                                                                                                                                                              |                                                                                                   |
| <p><b>WEEK 2</b> ☐ <b>45m Geology Lessons - Lesson 8</b></p> <p><i>Identification</i></p> <p>→ <b>RESEARCH</b></p> <p>Google your state's Geological Survey to see if they have any online resources that would be helpful to you, or check out the USGS maps online. If an office is nearby, they may have some maps or other resources you could pick up. Also, see if they will answer any of your questions about your local geology. If not, see if they know someone you could talk to.</p> <p>∞ Website Link: USGS Online Resources</p> <p>→ <b>PREP FOR FIELD DAY</b></p> <ul style="list-style-type: none"> <li>• Decide where you will explore on your field day.</li> <li>• Check the weather and confirm when you will go.</li> <li>• Collect any tools or materials that you might want to take with you.</li> </ul> |                                                                                                   |
| <p><b>WEEK 2</b> ☐ <b>45m Geology Lessons - Lesson 9</b></p> <p><i>Geology Literature</i></p> <div data-bbox="289 1770 989 1816" style="border: 1px solid #ccc; padding: 2px;"> <p>☐ Materials: Annals of the Former World</p> </div> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b></p> <p>~ 20 pages</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                   |



## Term 1

### WEEK 2 ☐ 45m Geology Lessons - Lesson 10

*Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 2 ☐ 60m Geology Labs - Lesson 2

#### → OBSERVE

Continue observations in the field. Sketch, describe, and document your local geology.

### WEEK 3 ☐ 45m Geology Lessons - Lesson 11

*How Geology Works*

☐ Materials: The Field Guide to Geology

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 3 ☐ 45m Geology Lessons - Lesson 12

*Practical Interpretation*

☐ Materials: Roadside Geology & Notebook

#### → READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.

### WEEK 3 ☐ 45m Geology Lessons - Lesson 13

*Identification*

#### → RESEARCH

Spend some more time considering the various geological maps. They can take some getting used to, so be patient!

∞ Website Link: USGS Online Resources

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.



## Term 1

### WEEK 3 45m Geology Lessons - Lesson 14

*Geology Literature*

Materials: Annals of the Former World

→ **READ, NARRATE, & DISCUSS**

~ 20 pages

### WEEK 3 45m Geology Lessons - Lesson 15

*Flex Day/Current Events*

Materials: Selected geology literature, lab/field supplies, or internet access

→ **NOTE**

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 3 60m Geology Labs - Lesson 3

→ **OBSERVE**

Continue observations in the field. Sketch, describe, and document your local geology.

### WEEK 4 45m Geology Lessons - Lesson 16

*How Geology Works*

Materials: The Field Guide to Geology

→ **READ, NARRATE, & DISCUSS**

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

#### • **PLAN WEEKLY**

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 4 45m Geology Lessons - Lesson 17

*Practical Interpretation*

Materials: Roadside Geology

→ **READ, NARRATE, & DISCUSS**

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.



## Term 1

### WEEK 4 ☐ 45m Geology Lessons - Lesson 18

#### *Identification*

☐ Materials: rock/mineral/fossil guides, painting tools of your choice & Nature Notebook

#### → RESEARCH

Start trying to identify any rocks, minerals, or fossils that you have picked up locally. Be sure to paint some of your rock finds in your Nature Notebook as you go through the term.

∞ Video Link: How to Draw Rocks

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 4 ☐ 45m Geology Lessons - Lesson 19

#### *Geology Literature*

☐ Materials: Annals of the Former World

#### → READ, NARRATE, & DISCUSS

~ 20 pages

### WEEK 4 ☐ 45m Geology Lessons - Lesson 20

#### *Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 4 ☐ 60m Geology Labs - Lesson 4

#### → OBSERVE

When you are driving around, start noticing where there are exposed rocks in your landscape. Take note of road beds that had to be cut out of the ground. Make observations of what you are seeing in your notebook. Include sketches when you are able. If there are any rivers, creeks, streams, etc. nearby, observe what sort of soil/rock has been exposed in the banks.



## Term 1

### WEEK 5 ☐ 45m Geology Lessons - Lesson 21

*How Geology Works*

☐ Materials: The Field Guide to Geology

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 5 ☐ 45m Geology Lessons - Lesson 22

*Practical Interpretation*

☐ Materials: Roadside Geology & Notebook

#### → READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.

### WEEK 5 ☐ 45m Geology Lessons - Lesson 23

*Identification*

☐ Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 5 ☐ 45m Geology Lessons - Lesson 24

*Geology Literature*

☐ Materials: Annals of the Former World

#### → READ, NARRATE, & DISCUSS

~ 20 pages



## Term 1

### WEEK 5 ☐ 45m Geology Lessons - Lesson 25

*Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 5 ☐ 60m Geology Labs - Lesson 5

#### → OBSERVE

Continue observations in the field. Sketch, describe, and document your local geology.

### WEEK 6 ☐ 45m Geology Lessons - Lesson 26

*How Geology Works*

☐ Materials: The Field Guide to Geology

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 6 ☐ 45m Geology Lessons - Lesson 27

*Practical Interpretation*

☐ Materials: Roadside Geology

#### → READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.

### WEEK 6 ☐ 45m Geology Lessons - Lesson 28

*Identification*

☐ Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

#### → PREP FOR FIELD DAY



## Term 1

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 6 45m Geology Lessons - Lesson 29

*Geology Literature*

Materials: Annals of the Former World

#### → READ, NARRATE, & DISCUSS

~ 20 pages

### WEEK 6 45m Geology Lessons - Lesson 30

*Flex Day/Current Events*

Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 6 60m Geology Labs - Lesson 6

#### → OBSERVE

Continue observations in the field. Sketch, describe, and document your local geology.

### WEEK 7 45m Geology Lessons - Lesson 31

*How Geology Works*

Materials: The Field Guide to Geology

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 7 45m Geology Lessons - Lesson 32

*Practical Interpretation*

Materials: Roadside Geology & Notebook

#### → READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.





## Term 1

### WEEK 7 ☐ 45m Geology Lessons - Lesson 33

#### *Identification*

☐ Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 7 ☐ 45m Geology Lessons - Lesson 34

#### *Geology Literature*

☐ Materials: Annals of the Former World

#### → READ, NARRATE, & DISCUSS

~ 20 pages

### WEEK 7 ☐ 45m Geology Lessons - Lesson 35

#### *Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 7 ☐ 60m Geology Labs - Lesson 7

#### → OBSERVE

Continue observations in the field. Sketch, describe, and document your local geology.

### WEEK 8 ☐ 45m Geology Lessons - Lesson 36

#### *How Geology Works*

☐ Materials: The Field Guide to Geology

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 1

Copy any diagrams or images that are helpful.

### WEEK 8 45m Geology Lessons - Lesson 37

*Practical Interpretation*

Materials: Roadside Geology & Notebook

#### → READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.

### WEEK 8 45m Geology Lessons - Lesson 38

*Identification*

Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 8 45m Geology Lessons - Lesson 39

*Geology Literature*

Materials: Annals of the Former World

#### → READ, NARRATE, & DISCUSS

~ 20 pages

### WEEK 8 45m Geology Lessons - Lesson 40

*Flex Day/Current Events*

Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.



## Term 1

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| <p><b>WEEK 8</b> ☐ <b>60m Geology Labs - Lesson 8</b></p> <p>→ <b>OBSERVE</b></p> <p>Continue observations in the field. Sketch, describe, and document your local geology.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                   |
| <p><b>WEEK 9</b> ☐ <b>45m Geology Lessons - Lesson 41</b></p> <p><i>How Geology Works</i></p> <div data-bbox="289 541 989 588" style="border: 1px solid #ccc; padding: 5px;"> <p>☐ Materials: The Field Guide to Geology</p> </div> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b></p> <p>Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>• <b>PLAN WEEKLY</b></p> <p>☐ nature walk - record observations</p> <p>☐ science free read</p> |
| <p><b>WEEK 9</b> ☐ <b>45m Geology Lessons - Lesson 42</b></p> <p><i>Practical Interpretation</i></p> <div data-bbox="289 835 989 882" style="border: 1px solid #ccc; padding: 5px;"> <p>☐ Materials: Roadside Geology &amp; notebook</p> </div> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b></p> <p>Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.</p>                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                   |
| <p><b>WEEK 9</b> ☐ <b>45m Geology Lessons - Lesson 43</b></p> <p><i>Identification</i></p> <div data-bbox="289 1161 989 1207" style="border: 1px solid #ccc; padding: 5px;"> <p>☐ Materials: rock/mineral/fossil guides</p> </div> <p>→ <b>RESEARCH</b></p> <p>Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.</p> <p>→ <b>PREP FOR FIELD DAY</b></p> <ul style="list-style-type: none"> <li>• Decide where you will explore on your field day.</li> <li>• Check the weather and confirm when you will go.</li> <li>• Collect any tools or materials that you might want to take with you.</li> </ul> |                                                                                                   |
| <p><b>WEEK 9</b> ☐ <b>45m Geology Lessons - Lesson 44</b></p> <p><i>Geology Literature</i></p> <div data-bbox="289 1728 989 1774" style="border: 1px solid #ccc; padding: 5px;"> <p>☐ Materials: Annals of the Former World</p> </div> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b></p> <p>~ 20 pages</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                   |



## Term 1

### WEEK 9 ☐ 45m Geology Lessons - Lesson 45

#### *Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 9 ☐ 60m Geology Labs - Lesson 9

#### → OBSERVE

Continue observations in the field. Sketch, describe, and document your local geology.

### WEEK 10 ☐ 45m Geology Lessons - Lesson 46

#### *How Geology Works*

☐ Materials: The Field Guide to Geology

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 10 ☐ 45m Geology Lessons - Lesson 47

#### *Practical Interpretation*

☐ Materials: Roadside Geology & Notebook

#### → READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.

### WEEK 10 ☐ 45m Geology Lessons - Lesson 48

#### *Identification*

☐ Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

#### → PREP FOR FIELD DAY



## Term 1

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 10 ☐ 45m Geology Lessons - Lesson 49

#### *Geology Literature*

☐ Materials: Annals of the Former World

#### → READ, NARRATE, & DISCUSS

~ 20 pages

### WEEK 10 ☐ 45m Geology Lessons - Lesson 50

#### *Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 10 ☐ 60m Geology Labs - Lesson 10

#### → OBSERVE

Continue your local exploration, collection, and documentation.

### WEEK 11 ☐ 45m Geology Lessons - Lesson 51

#### *How Geology Works*

☐ Materials: The Field Guide to Geology

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 11 ☐ 45m Geology Lessons - Lesson 52

#### *Practical Interpretation*

☐ Materials: Roadside Geology & Notebook

#### → READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.



## Term 1

### WEEK 11 45m Geology Lessons - Lesson 53

*Identification*

Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally.

### WEEK 11 45m Geology Lessons - Lesson 54

*Geology Literature*

Materials: Annals of the Former World

#### → READ, NARRATE, & DISCUSS

~ 20 pages

### WEEK 11 45m Geology Lessons - Lesson 55

*Present What You've Learned*

#### → PREPARE

For one of your exam questions next week, you will be asked to give an informal presentation to your teacher on what you have discovered so far about your local geology and geological history. Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like. You will do a more formal presentation at the end of the year. This is more like an update on what you have discovered so far.

### WEEK 11 60m Geology Labs - Lesson 11

#### → OBSERVE

Continue your local exploration, collection, and documentation.

### WEEK 12 45m Geology Lessons - Lesson 56

*Exam Week*

#### → EXAMS

Answer question(s) related to course.

### WEEK 12 45m Geology Lessons - Lesson 57

*Exam Week*

#### → EXAMS

Answer question(s) related to course.

### WEEK 12 45m Geology Lessons - Lesson 58

*Exam Week*

#### → EXAMS

Answer question(s) related to course.



| Term 1                                                                                                                                                                              |  |
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| <div><div>WEEK 12</div><div><div></div>45m Geology Lessons - Lesson 59</div><div>Exam Week</div><div><div>→ EXAMS</div><div>Answer question(s) related to course.</div></div></div> |  |
| <div><div>WEEK 12</div><div><div></div>45m Geology Lessons - Lesson 60</div><div>Exam Week</div><div><div>→ EXAMS</div><div>Answer question(s) related to course.</div></div></div> |  |
| <div><div>WEEK 12</div><div><div></div>60m Geology Labs - Lesson 12</div><div>Exam Week</div><div><div>→ EXAMS</div><div>Answer question(s) related to course.</div></div></div>    |  |



## Term 2

### WEEK 1 ☐ 45m Geology Lessons - Lesson 61

*How Geology Works*

☐ Materials: The Field Guide to Geology

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 1 ☐ 45m Geology Lessons - Lesson 62

*Practical Interpretation*

☐ Materials: 101 American Geo-Sites You've Gotta See & Notebook

#### → INTRO

You read about your state or region last term; this term, we will expand the survey to the country's borders.

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 1 ☐ 45m Geology Lessons - Lesson 63

*Identification*

#### → RESEARCH

Where do you plan to visit this term? Is there a specific rock type or formation that you want to learn more about? Use the resources that you discovered last term to make a plan for this term. What questions about your local geology do you have? How will you find the answers?

- ∞ Link: Gator Girl Rocks
- ∞ Link: How to Find Rocks
- ∞ Link: Rock Clubs by State

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 1 ☐ 45m Geology Lessons - Lesson 64

*Geology Literature*

☐ Materials: Annals of the Former World

#### → READ, NARRATE, & DISCUSS





## Term 2

~ 20 pages

### WEEK 1 45m Geology Lessons - Lesson 65

*Flex Day/Current Events*

Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 1 60m Geology Labs - Lesson 13

#### → OBSERVE

Continue to use this time to explore your local geology. Continue sketching, describing, and documenting.

### WEEK 2 45m Geology Lessons - Lesson 66

*How Geology Works*

Materials: The Field Guide to Geology

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 2 45m Geology Lessons - Lesson 67

*Practical Interpretation*

Materials: 101 American Geo-Sites You've Gotta See & Notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 2 45m Geology Lessons - Lesson 68

*Identification*

Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your



## Term 2

area and that they have found helpful.

### → PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 2 ☐ 45m Geology Lessons - Lesson 69

*Geology Literature*

☐ Materials: Annals of the Former World

### → READ, NARRATE, & DISCUSS

~ 20 pages

### WEEK 2 ☐ 45m Geology Lessons - Lesson 70

*Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 2 ☐ 60m Geology Labs - Lesson 14

### → OBSERVE

Continue to use this time to explore your local geology. Continue building a rock, mineral, and/or fossil collection. Continue sketching, describing, and documenting.

### WEEK 3 ☐ 45m Geology Lessons - Lesson 71

*How Geology Works*

☐ Materials: The Field Guide to Geology

### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



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### WEEK 3 45m Geology Lessons - Lesson 72

#### *Practical Interpretation*

Materials: 101 American Geo-Sites You've Gotta See & Notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 3 45m Geology Lessons - Lesson 73

#### *Identification*

Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Consider taking part in a citizen science project on your field days:

∞ Link: Citizen Science Sand Project

∞ Link: Chronolog

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 3 45m Geology Lessons - Lesson 74

#### *Geology Literature*

Materials: Annals of the Former World

#### → READ, NARRATE, & DISCUSS

~ 20 pages

### WEEK 3 45m Geology Lessons - Lesson 75

#### *Flex Day/Current Events*

Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.



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### WEEK 3 ☐ 60m Geology Labs - Lesson 15

#### → OBSERVE

Continue to use this time to explore your local geology. Continue building a rock, mineral, and/or fossil collection. Continue sketching, describing, and documenting.

### WEEK 4 ☐ 45m Geology Lessons - Lesson 76

#### *How Geology Works*

☐ Materials: The Field Guide to Geology

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 4 ☐ 45m Geology Lessons - Lesson 77

#### *Practical Interpretation*

☐ Materials: 101 American Geo-Sites You've Gotta See & Notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 4 ☐ 45m Geology Lessons - Lesson 78

#### *Identification*

☐ Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. If you haven't looked at them under magnification yet, give it a try! Read the linked article for inspiration:  
∞ Article Link: Sand Grains

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 4 ☐ 45m Geology Lessons - Lesson 79

#### *Geology Literature*

☐ Materials: Annals of the Former World

#### → READ, NARRATE, & DISCUSS

~ 20 pages



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### WEEK 4 ☐ 45m Geology Lessons - Lesson 80

#### *Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 4 ☐ 60m Geology Labs - Lesson 16

#### → OBSERVE

Continue to notice where there are exposed rocks in your landscape. Take note of road beds that had to be cut out of the ground. Make observations of what you are seeing in your notebook. Include sketches when you are able. If there are any rivers, creeks, streams, etc. nearby, observe what sort of soil/rock has been exposed in the banks.

### WEEK 5 ☐ 45m Geology Lessons - Lesson 81

#### *How Geology Works*

☐ Materials: The Field Guide to Geology

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 5 ☐ 45m Geology Lessons - Lesson 82

#### *Practical Interpretation*

☐ Materials: 101 American Geo-Sites You've Gotta See & Notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 5 ☐ 45m Geology Lessons - Lesson 83

#### *Identification*

☐ Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and



## Term 2

direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

### → PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 5 ☐ 45m Geology Lessons - Lesson 84

*Geology Literature*

☐ Materials: Annals of the Former World

### → READ, NARRATE, & DISCUSS

~ 20 pages

### WEEK 5 ☐ 45m Geology Lessons - Lesson 85

*Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 5 ☐ 60m Geology Labs - Lesson 17

### → OBSERVE

Continue observations in the field. Sketch, describe, and document your local geology.

### WEEK 6 ☐ 45m Geology Lessons - Lesson 86

*How Geology Works*

☐ Materials: The Field Guide to Geology

### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



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### WEEK 6 45m Geology Lessons - Lesson 87

#### *Practical Interpretation*

Materials: 101 American Geo-Sites You've Gotta See & Notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 6 45m Geology Lessons - Lesson 88

#### *Identification*

Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Consider taking part in a citizen science project on your field days:

∞ Link: Citizen Science Sand Project

∞ Link: Chronolog

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.

- Check the weather and confirm when you will go.

- Collect any tools or materials that you might want to take with you.

### WEEK 6 45m Geology Lessons - Lesson 89

#### *Geology Literature*

Materials: Annals of the Former World

#### → READ, NARRATE, & DISCUSS

~ 20 pages

### WEEK 6 45m Geology Lessons - Lesson 90

#### *Flex Day/Current Events*

Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.



## Term 2

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>WEEK 6</b> ☐ <b>60m Geology Labs - Lesson 18</b><br/> <b>→ OBSERVE</b><br/>         Continue observations in the field. Sketch, describe, and document your local geology.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                |
| <p><b>WEEK 7</b> ☐ <b>45m Geology Lessons - Lesson 91</b><br/> <i>How Geology Works</i></p> <div data-bbox="292 541 989 588" style="border: 1px solid black; padding: 5px;"> <p>☐ Materials: The Field Guide to Geology</p> </div> <p><b>→ READ, NARRATE, &amp; DISCUSS</b><br/>         Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>• <b>PLAN WEEKLY</b></p> <ul style="list-style-type: none"> <li>☐ nature walk - record observations</li> <li>☐ science free read</li> </ul> |
| <p><b>WEEK 7</b> ☐ <b>45m Geology Lessons - Lesson 92</b><br/> <i>Practical Interpretation</i></p> <div data-bbox="292 835 989 915" style="border: 1px solid black; padding: 5px;"> <p>☐ Materials: 101 American Geo-Sites You've Gotta See &amp; Notebook</p> </div> <p><b>→ READ, NARRATE, &amp; DISCUSS</b><br/>         ~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.</p>                                                                                                                                                                                                                                                                                      |                                                                                                                                                |
| <p><b>WEEK 7</b> ☐ <b>45m Geology Lessons - Lesson 93</b><br/> <i>Identification</i></p> <div data-bbox="292 1228 989 1276" style="border: 1px solid black; padding: 5px;"> <p>☐ Materials: rock/mineral/fossil guides</p> </div> <p><b>→ RESEARCH</b><br/>         Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.</p> <p><b>→ PREP FOR FIELD DAY</b></p> <ul style="list-style-type: none"> <li>• Decide where you will explore on your field day.</li> <li>• Check the weather and confirm when you will go.</li> <li>• Collect any tools or materials that you might want to take with you.</li> </ul> |                                                                                                                                                |
| <p><b>WEEK 7</b> ☐ <b>45m Geology Lessons - Lesson 94</b><br/> <i>Geology Literature</i></p> <div data-bbox="292 1791 989 1839" style="border: 1px solid black; padding: 5px;"> <p>☐ Materials: Annals of the Former World</p> </div> <p><b>→ READ, NARRATE, &amp; DISCUSS</b><br/>         ~ 20 pages</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                |





## Term 2

### WEEK 7 ☐ 45m Geology Lessons - Lesson 95

*Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 7 ☐ 60m Geology Labs - Lesson 19

#### → OBSERVE

Continue observations in the field. Sketch, describe, and document your local geology.

### WEEK 8 ☐ 45m Geology Lessons - Lesson 96

*How Geology Works*

☐ Materials: The Field Guide to Geology

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 8 ☐ 45m Geology Lessons - Lesson 97

*Practical Interpretation*

☐ Materials: 101 American Geo-Sites You've Gotta See & Notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 8 ☐ 45m Geology Lessons - Lesson 98

*Identification*

☐ Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.



## Term 2

### → PREP

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 8 45m Geology Lessons - Lesson 99

#### *Geology Literature*

Materials: Annals of the Former World

### → READ, NARRATE, & DISCUSS

~ 20 pages

### WEEK 8 45m Geology Lessons - Lesson 100

#### *Flex Day/Current Events*

Materials: Selected geology literature, lab/field supplies, or internet access

### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 8 60m Geology Labs - Lesson 20

### → OBSERVE

Continue your local exploration, collection, and documentation.

### WEEK 9 45m Geology Lessons - Lesson 101

#### *How Geology Works*

Materials: The Field Guide to Geology

### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 9 45m Geology Lessons - Lesson 102

#### *Practical Interpretation*

Materials: 101 American Geo-Sites You've Gotta See & Notebook

### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend



## Term 2

with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 9 45m Geology Lessons - Lesson 103

#### Identification

Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 9 45m Geology Lessons - Lesson 104

#### Geology Literature

Materials: Annals of the Former World

#### → READ, NARRATE, & DISCUSS

~ 20 pages

### WEEK 9 45m Geology Lessons - Lesson 105

#### Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 9 60m Geology Labs - Lesson 21

#### → OBSERVE

Continue your local exploration, collection, and documentation.



## Term 2

### WEEK 10 ☐ 45m Geology Lessons - Lesson 106

*How Geology Works*

☐ Materials: The Field Guide to Geology

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you.  
Copy any diagrams or images that are helpful.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 10 ☐ 45m Geology Lessons - Lesson 107

*Practical Interpretation*

☐ Materials: 101 American Geo-Sites You've Gotta See & Notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 10 ☐ 45m Geology Lessons - Lesson 108

*Identification*

☐ Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 10 ☐ 45m Geology Lessons - Lesson 109

*Geology Literature*

☐ Materials: Annals of the Former World

#### → READ, NARRATE, & DISCUSS

~ 20 pages



## Term 2

### WEEK 10 ☐ 45m Geology Lessons - Lesson 110

*Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

### WEEK 10 ☐ 60m Geology Labs - Lesson 22

#### → OBSERVE

Continue your local exploration, collection, and documentation.

### WEEK 11 ☐ 45m Geology Lessons - Lesson 111

*How Geology Works*

☐ Materials: The Field Guide to Geology

#### → READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 11 ☐ 45m Geology Lessons - Lesson 112

*Practical Interpretation*

☐ Materials: 101 American Geo-Sites You've Gotta See & Notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 11 ☐ 45m Geology Lessons - Lesson 113

*Identification*

☐ Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally.



## Term 2

### WEEK 11 45m Geology Lessons - Lesson 114

*Geology Literature*

Materials: Annals of the Former World

→ **READ, NARRATE, & DISCUSS**

~ 20 pages

### WEEK 11 45m Geology Lessons - Lesson 115

*Present What You've Learned*

→ **PREPARE**

For one of your exam questions next week, you will be asked to give another informal presentation to your teacher on what you have discovered during this second term about your local geology and geological history. Use this time to gather your thoughts and think through how you want to do that. You can be as creative as you would like. You will do a more formal presentation at the end of the year. This is more like an update on what you have discovered so far.

### WEEK 11 60m Geology Labs - Lesson 23

→ **OBSERVE**

Continue your local exploration, collection, and documentation.

### WEEK 12 45m Geology Lessons - Lesson 116

*Exam Week*

→ **EXAMS**

Answer question(s) related to course.

### WEEK 12 45m Geology Lessons - Lesson 117

*Exam Week*

→ **EXAMS**

Answer question(s) related to course.

### WEEK 12 45m Geology Lessons - Lesson 118

*Exam Week*

→ **EXAMS**

Answer question(s) related to course.

### WEEK 12 45m Geology Lessons - Lesson 119

*Exam Week*

→ **EXAMS**

Answer question(s) related to course.



| Term 2                                                                                                                                                                               |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div><div>WEEK 12</div><div><div></div>45m Geology Lessons - Lesson 120</div><div>Exam Week</div><div><div>→ EXAMS</div><div>Answer question(s) related to course.</div></div></div> |  |
| <div><div>WEEK 12</div><div><div></div>60m Geology Labs - Lesson 24</div><div>Exam Week</div><div><div>→ EXAMS</div><div>Answer question(s) related to course.</div></div></div>     |  |



## Term 3

### WEEK 1 45m Geology Lessons - Lesson 121

*How Geology Works*

Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

#### • PLAN WEEKLY

- nature walk - record observations
- science free read

### WEEK 1 45m Geology Lessons - Lesson 122

*Practical Interpretation*

Materials: 101 American Geo-Sites You've Gotta See & notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 1 45m Geology Lessons - Lesson 123

*Identification*

#### → RESEARCH

Where do you plan to visit this term? Is there a specific rock type or formation that you want to learn more about? Use the resources that you discovered last term to make a plan for this term. What questions about your local geology do you have? How will you find the answers?

- ∞ Link: Gator Girl Rocks
- ∞ Link: How to Find Rocks
- ∞ Link: Rock Clubs by State

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 1 45m Geology Lessons - Lesson 124

*How Geology Works*

Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.





## Term 3

### WEEK 1 ☐ 45m Geology Lessons - Lesson 125

*Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

OR

#### → VIEW & NARRATE

You might check out this video introduction to rocks and the geology of the Lake Superior region.

∞ Video Link: Wolf Ridge, MN

### WEEK 1 ☐ 60m Geology Labs - Lesson 25

#### → OBSERVE

Continue to use this time to explore your local geology.

### WEEK 2 ☐ 45m Geology Lessons - Lesson 126

*How Geology Works*

☐ Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 2 ☐ 45m Geology Lessons - Lesson 127

*Practical Interpretation*

☐ Materials: 101 American Geo-Sites You've Gotta See & notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 2 ☐ 45m Geology Lessons - Lesson 128

*Identification*

☐ Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up



## Term 3

locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

### → PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

## WEEK 2 45m Geology Lessons - Lesson 129

*How Geology Works*

Materials: The Story of the Earth in 25 Rocks

### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

## WEEK 2 45m Geology Lessons - Lesson 130

*Flex Day/Current Events*

Materials: Selected geology literature, lab/field supplies, or internet access

### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

OR

### → VIEW & NARRATE

Watch this video about the 1906 earthquake. You may remember a map in your *All Over the Map* geography book about the boats that sank in the harbor he references here. Take a look!

∞ Video Link: The Gold Rush and the 1906 Earthquake

## WEEK 2 60m Geology Labs - Lesson 26

### → OBSERVE

Continue to use this time to explore your local geology.



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### WEEK 3 45m Geology Lessons - Lesson 131

*How Geology Works*

Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 3 45m Geology Lessons - Lesson 132

*Practical Interpretation*

Materials: 101 American Geo-Sites You've Gotta See & notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 3 45m Geology Lessons - Lesson 133

*Identification*

Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Consider taking part in a citizen science project on your field days:

- ∞ Link: Citizen Science Sand Project
- ∞ Link: Chronolog

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 3 45m Geology Lessons - Lesson 134

*How Geology Works*

Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

### WEEK 3 45m Geology Lessons - Lesson 135

*Flex Day/Current Events*

Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology



## Term 3

books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

OR

### → VIEW & NARRATE

Choose a few of these videos about our national parks.

∞ Video Link: National Parks Geology

### WEEK 3 ☐ 60m Geology Labs - Lesson 27

#### → OBSERVE

Continue to use this time to explore your local geology.

### WEEK 4 ☐ 45m Geology Lessons - Lesson 136

*How Geology Works*

☐ Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 4 ☐ 45m Geology Lessons - Lesson 137

*Practical Interpretation*

☐ Materials: 101 American Geo-Sites You've Gotta See & notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 4 ☐ 45m Geology Lessons - Lesson 138

*Identification*

☐ Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.



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### WEEK 4 ☐ 45m Geology Lessons - Lesson 139

*How Geology Works*

☐ Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

### WEEK 4 ☐ 45m Geology Lessons - Lesson 140

*Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

OR

#### → VIEW & NARRATE

Choose a few of these videos about our national parks.

∞ Video Link: National Parks Geology

### WEEK 4 ☐ 60m Geology Labs - Lesson 28

#### → OBSERVE

Continue to notice where there are exposed rocks in your landscape. Take note of road beds that had to be cut out of the ground. Make observations of what you are seeing in your notebook. Include sketches when you are able. If there are any rivers, creeks, streams, etc. nearby, observe what sort of soil/rock has been exposed in the banks.

### WEEK 5 ☐ 45m Geology Lessons - Lesson 141

*How Geology Works*

☐ Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 5 ☐ 45m Geology Lessons - Lesson 142

*Practical Interpretation*

☐ Materials: 101 American Geo-Sites You've Gotta See & notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend



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with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 5 45m Geology Lessons - Lesson 143

#### Identification

Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 5 45m Geology Lessons - Lesson 144

#### How Geology Works

Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

### WEEK 5 45m Geology Lessons - Lesson 145

#### Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

OR

#### → VIEW & NARRATE

Spend time exploring other geology articles at this site after watching this one about Marie Tharp!

∞ Article Link: Marie Tharp's Groundbreaking Maps



## Term 3

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <p><b>WEEK 5</b> ☐ <b>60m Geology Labs - Lesson 29</b><br/> <b>→ OBSERVE</b><br/>         Continue observations in the field. Sketch, describe, and document your local geology.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                   |
| <p><b>WEEK 6</b> ☐ <b>45m Geology Lessons - Lesson 146</b><br/> <i>How Geology Works</i></p> <div data-bbox="289 541 989 588" style="border: 1px solid black; padding: 2px;"> <p>☐ Materials: The Story of the Earth in 25 Rocks</p> </div> <p><b>→ READ, NARRATE, &amp; DISCUSS</b><br/>         2-3 chapters per week; note that you have 2 lesson days for this book.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>• <b>PLAN WEEKLY</b></p> <p>☐ nature walk - record observations</p> <p>☐ science free read</p> |
| <p><b>WEEK 6</b> ☐ <b>45m Geology Lessons - Lesson 147</b><br/> <i>Practical Interpretation</i></p> <div data-bbox="289 804 989 848" style="border: 1px solid black; padding: 2px;"> <p>☐ Materials: 101 American Geo-Sites You've Gotta See &amp; notebook</p> </div> <p><b>→ READ, NARRATE, &amp; DISCUSS</b><br/>         ~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.</p>                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                   |
| <p><b>WEEK 6</b> ☐ <b>45m Geology Lessons - Lesson 148</b><br/> <i>Identification</i></p> <div data-bbox="289 1161 989 1207" style="border: 1px solid black; padding: 2px;"> <p>☐ Materials: rock/mineral/fossil guides</p> </div> <p><b>→ RESEARCH</b><br/>         Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.</p> <p><b>→ PREP FOR FIELD DAY</b></p> <ul style="list-style-type: none"> <li>• Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.</li> <li>• Check the weather and confirm when you will go.</li> <li>• Collect any tools or materials that you might want to take with you.</li> </ul> |                                                                                                   |
| <p><b>WEEK 6</b> ☐ <b>45m Geology Lessons - Lesson 149</b><br/> <i>How Geology Works</i></p> <div data-bbox="289 1791 989 1837" style="border: 1px solid black; padding: 2px;"> <p>☐ Materials: The Story of the Earth in 25 Rocks</p> </div> <p><b>→ READ, NARRATE, &amp; DISCUSS</b><br/>         2-3 chapters per week; note that you have 2 lesson days for this book.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                   |



## Term 3

### WEEK 6 ☐ 45m Geology Lessons - Lesson 150

*Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

OR

#### → VIEW & NARRATE

Watch this video about geology and rockfalls.

∞ Video Link: Rockfalls at Yosemite

### WEEK 6 ☐ 60m Geology Labs - Lesson 30

#### → OBSERVE

Continue observations in the field. Sketch, describe, and document your local geology.

### WEEK 7 ☐ 45m Geology Lessons - Lesson 151

*How Geology Works*

☐ Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 7 ☐ 45m Geology Lessons - Lesson 152

*Practical Interpretation*

☐ Materials: 101 American Geo-Sites You've Gotta See & notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 7 ☐ 45m Geology Lessons - Lesson 153

*Identification*

☐ Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up





## Term 3

locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 7 ☐ 45m Geology Lessons - Lesson 154

*How Geology Works*

☐ Materials: The Story of the Earth in 25 Rocks

### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

### WEEK 7 ☐ 45m Geology Lessons - Lesson 155

*Flex Day/Current Events*

☐ Materials: Selected geology literature, lab/field supplies, or internet access

### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

OR

### → LISTEN/VIEW

Choose some podcast/video episodes:

∞ Audio Link: Geo Podcast

∞ Video Link: Geology Video Playlist

### WEEK 7 ☐ 60m Geology Labs - Lesson 31

### → OBSERVE

Continue your local exploration, collection, and documentation.

### WEEK 8 ☐ 45m Geology Lessons - Lesson 156

*How Geology Works*

☐ Materials: The Story of the Earth in 25 Rocks

### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 3

### WEEK 8 45m Geology Lessons - Lesson 157

#### *Practical Interpretation*

Materials: 101 American Geo-Sites You've Gotta See & notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 8 45m Geology Lessons - Lesson 158

#### *Identification*

Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 8 45m Geology Lessons - Lesson 159

#### *How Geology Works*

Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

### WEEK 8 45m Geology Lessons - Lesson 160

#### *Flex Day/Current Events*

Materials: Selected geology literature, lab/field supplies, or internet access

#### → NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

OR



## Term 3

### → VIEW & NARRATE

Watch these fascinating videos about geology, music, and art! We live in an amazing world! Art and music help us to experience and see things in new ways. The last video is a bit long, so finish it at another time if you are interested.

- ∞ Video Link: Ringing Rocks
- ∞ Video Link: Balancing Rocks 1
- ∞ Video Link: Balancing Rocks 2
- ∞ Video Link: Andy Goldsworthy

### WEEK 8 ☐ 60m Geology Labs - Lesson 32

#### → OBSERVE

Continue your local exploration, collection, and documentation.

### WEEK 9 ☐ 45m Geology Lessons - Lesson 161

*How Geology Works*

☐ Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 9 ☐ 45m Geology Lessons - Lesson 162

*Practical Interpretation*

☐ Materials: 101 American Geo-Sites You've Gotta See & notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 9 ☐ 45m Geology Lessons - Lesson 163

*Identification*

☐ Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.



## Term 3

### WEEK 9 ☐ 45m Geology Lessons - Lesson 164

*How Geology Works*

☐ Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

### WEEK 9 ☐ 45m Geology Lessons - Lesson 165

*Present What You've Learned*

#### → PLAN

Make a plan for how you will take all the information you have researched on your local area: fieldwork, experiments, photos, illustrations, drawings, maps, and specimens, and prepare a presentation about your local geology and its history, along with any other places you visited.

You might type up notes and give your presentation as a talk; you might want to create a PowerPoint or Google Slides presentation. Or, you might choose to design something in Canva or another visual tool. You might like to create a display that uses maps and charts with your specimens laid out and labeled. You might want to design an infographic that uses your specimens to describe your local area. You decide. Be as creative as you would like. You can include poetry or music that you write, poetry and music that you find, and/or incorporate paintings, drawings, or illustrations.

You will have three weeks to work on it. Then, you will give your presentation to a live audience during exam week.

### WEEK 9 ☐ 60m Geology Labs - Lesson 33

#### → OBSERVE

Continue your local exploration, collection, and documentation.

### WEEK 10 ☐ 45m Geology Lessons - Lesson 166

*How Geology Works*

☐ Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 10 ☐ 45m Geology Lessons - Lesson 167

*Practical Interpretation*

☐ Materials: 101 American Geo-Sites You've Gotta See & notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend



## Term 3

with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 10 45m Geology Lessons - Lesson 168

#### Identification

Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

#### → PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

### WEEK 10 45m Geology Lessons - Lesson 169

#### How Geology Works

Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

### WEEK 10 45m Geology Lessons - Lesson 170

#### Present What You've Learned

#### → PREPARE

Continue to prepare your presentation.

### WEEK 10 60m Geology Labs - Lesson 34

#### → OBSERVE

Continue your local exploration, collection, and documentation.

### WEEK 11 45m Geology Lessons - Lesson 171

#### How Geology Works

Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 3

### WEEK 11 ☐ 45m Geology Lessons - Lesson 172

*Practical Interpretation*

☐ Materials: 101 American Geo-Sites You've Gotta See & notebook

#### → READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

### WEEK 11 ☐ 45m Geology Lessons - Lesson 173

*Identification*

☐ Materials: rock/mineral/fossil guides

#### → RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally.

### WEEK 11 ☐ 45m Geology Lessons - Lesson 174

*How Geology Works*

☐ Materials: The Story of the Earth in 25 Rocks

#### → READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

### WEEK 11 ☐ 45m Geology Lessons - Lesson 175

*Present What You've Learned*

#### → PREPARE

Continue to prepare your presentation. Enjoy giving it next week to a live audience!

### WEEK 11 ☐ 60m Geology Labs - Lesson 35

#### → OBSERVE

Continue your local exploration, collection, and documentation.

### WEEK 12 ☐ 45m Geology Lessons - Lesson 176

*Exam Week*

#### → EXAMS

Answer question(s) related to course.

### WEEK 12 ☐ 45m Geology Lessons - Lesson 177

*Exam Week*

#### → EXAMS

Answer question(s) related to course.



| Term 3                                                                                                                                                                                |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div><div>WEEK 12</div><div><div></div>45m Geology Lessons - Lesson 178</div><div>Exam Week</div></div> <div><div>→ EXAMS</div><div>Answer question(s) related to course.</div></div> |  |
| <div><div>WEEK 12</div><div><div></div>45m Geology Lessons - Lesson 179</div><div>Exam Week</div></div> <div><div>→ EXAMS</div><div>Answer question(s) related to course.</div></div> |  |
| <div><div>WEEK 12</div><div><div></div>45m Geology Lessons - Lesson 180</div><div>Exam Week</div></div> <div><div>→ EXAMS</div><div>Answer question(s) related to course.</div></div> |  |
| <div><div>WEEK 12</div><div><div></div>60m Geology Labs - Lesson 36</div><div>Exam Week</div></div> <div><div>→ EXAMS</div><div>Answer question(s) related to course.</div></div>     |  |

# Science: Geology

## Examination

### Term 1

#### Geology Lessons

- Field Guide to Geology: Explain in what ways the earth's crust changes.
- Field Guide to Geology: Describe different ways that rocks are formed through the changes in the earth's crust.
- Explain how you identified a found rock this term. OR Tell about a geology current event.
- Annals of a Former World: Explain some aspect of geology that McPhee has brought to life for you.
- Explain what you have discovered so far about your local geology and geological history. Share your notebook.

### Term 2

#### Geology Lessons

- Field Guide to Geology: Describe how water shapes the land through rivers OR seas OR ice.
- Field Guide to Geology: Explain how relative dating works.
- Explain how you identified a found rock this term. OR Tell about a geology current event.
- Annals of a Former World: Explain some aspect of geology that McPhee has brought to life for you.
- Tell what you have discovered this term about your local geology and geological history. Share your notebook.

### Term 3

#### Geology Lessons

- Tell the story of one of the rocks from The Story of the Earth in 25 Rocks.
- The Story of the Earth in 25 Rocks: Explain how a geologist is like a detective.
- Explain how you identified a found rock this term. OR Tell about a geology current event.
- Tell about a geological site that you read about that you would like to visit someday. Where is it? What is interesting about it? What would you hope to see or learn?
- Give a presentation on your local geology and geological history. Share your notebook.